

QA Dashboard

AI Literacy for the Modern Workforce

Quality assurance on this project adapted Hardman's AI Evals framework to an L&D context: a specify-then-verify loop with binary pass/fail judgments.

When the practitioner's core job shifts from authoring content to evaluating AI-generated content, the critical skills become specification (defining "good" precisely enough for AI to produce it) and verification (reliably detecting when output fails that bar). The method is a binary pass/fail cycle: select an output type, collect samples, define checks, score, diagnose root cause (specification failure vs. generalization failure), revise the specification, re-score, and repeat on a schedule.

Applied across four output categories and six data files, it ran as ten structured verification passes, a triple-validation protocol for the data files, and an eight-check capability framework, all recorded below. Everything the passes surfaced was resolved before or during the build.

TEN VERIFICATION PASSES

ACTIVITY	METHOD	SCOPE	SAMPLE	RESULT
Capability check logs (M1–M4)	Inline pass/fail checks during authoring	Narrative, scenario stems, feedback, KC items	4 modules	All passed; outstanding items resolved before build handoff.
Knowledge-check evaluation	Binary checks: scenario-based, application-level, plausible distractors, consequence feedback, no cueing	16 KC items across 4 modules	16 items	KC-1.1–1.3 revised for distractor balance; KC-2.3/2.4 for forward-reference; KC-4.4 Option C for length parity.
Practice-activity traceability audit	Each P-activity vs. action-map and content D-codes	12 practice activities (P1–P12)	12 items	All 12 match across both sources in behavior targets and descriptions.
KC-objective traceability	Each KC stem vs. its referenced performance objective	16 KCs against 17 objectives	16 items	All 16 correctly assess their referenced objectives.
Data-file triple validation	Schema, value verification vs. sources, cross-file consistency	6 JSON files	6 × 3	All validated; \$55 → \$54 and citation page-range corrections applied.
Cross-cutting integration check (1)	Event model, state shape, completion logic, consumer mapping	5 component specs + 6 data files	28 edits	95 event keys cataloged; 37+ state keys reconciled; master event-key table produced.
Cross-cutting integration check (2)	Content flow, terminology, tone, D-code trace, color batch fix	4 content docs + 2 planning + 4 specs	6 tasks	All clean; six O3a alignment-matrix discrepancies corrected; two stale markers flagged.
JSX artifact audit	Reconcile interactive diagrams vs. the built codebase	2 JSX files against src/	18 found	5 major, 13 minor found and resolved; downstream doc fixes applied.

ACTIVITY	METHOD	SCOPE	SAMPLE	RESULT
Citation verification pass	Every inline statistic vs. reference–quotes.md and source papers	21 statistics across 4 modules	21 items	One known discrepancy (81% vs 84% median savings) confirmed resolved per the Figure 6 methodology.
Reference item (R1–R7) design review	PDF assets vs. design system; content accuracy and register	7 reference items at 10 placements	7 items	All 7 redesigned in the project design system; contextual placement verified.

TRIPLE-VALIDATION PROTOCOL

Three dedicated passes were run on every data file before it reached a chart:

PASS	NAME	DETAIL
1	Schema validation	Field names, data types, required vs. optional fields, and structural consistency across array entries.
2	Value verification	Every numeric value and quoted string traced to a specific page, table, or figure in the source paper, cross-checked against reference–quotes.md.
3	Cross-file consistency	Values shared across files (e.g., the 81% median–savings figure) verified for consistency; each file’s consuming components confirmed against the specs.

Six validated JSON files

FILE	SOURCE	CONTENT	STATUS
augmentation-automation.json	Handa et al.	Occupation/task breakdowns; 57%/43% augmentation–automation split	PASS
productivity-gains.json	Tamkin & McCrory	Task time reductions; 81% median savings (Fig. 6 caption, p. 13)	PASS · 2 FIXES
geographic-adoption.json	Appel, McCrory & Tamkin	Country-level usage rates; institutional-inertia indicators	PASS
wef-skill-demand.json	WEF Future of Jobs 2025	Fastest-growing/declining skills; employer transformation expectations	PASS
interviewer-findings.json	Anthropic Interviewer	86% time savings; 65% satisfaction; 69% stigma; 48% career transition	PASS
adoption-trends.json	Multiple sources	Longitudinal adoption rates; interaction-mode temporal data	PASS

EIGHT-CHECK CAPABILITY FRAMEWORK

Each knowledge-check item was scored against eight binary checks, tagged either as a **capability** the item must demonstrate or a **risk** it must avoid:

CHECK	TYPE	CRITERION
Scenario authenticity	CAPABILITY	Does the stem present a recognizable workplace situation with a job title, context, and decision point?
Application-level cognitive demand	CAPABILITY	Does the item test applying a concept to a novel situation rather than recalling a definition?
Distractor plausibility	RISK	Are all non-preferred options actions a reasonable professional might consider? (No trick or absurd choices.)
Correct-answer cueing absence	RISK	Is the preferred option free of length, specificity, or hedging cues that distinguish it without content knowledge?
Consequence-based feedback	CAPABILITY	Does each option branch lead to a realistic observed outcome rather than a "Correct" / "Try again" label?
Source fidelity	RISK	Does every cited statistic match its source paper (checked against reference-quotes.md)?
Register consistency	CAPABILITY	Does the text hold the "knowledgeable peer" second-person register without slipping into lecture or compliance tone?
Forward-reference avoidance	RISK	Does the item avoid requiring knowledge from a module the learner has not yet completed?

THE SINGLE-EVALUATOR LIMIT

The structured passes above were run by the same person who authored the content, so specification failures are hardest to detect where the specifier and the evaluator share a mental model. An independent review of both the code and the content by a different model (Claude Fable) provided a second pass that did not share that mental model and surfaced issues that were then fixed. That does not replace what an organizational deployment should add: the evaluation role separated from authoring, with inter-rater reliability established for the check framework. But it is more than a single set of eyes, and the binary pass/fail format is designed to transfer: it is specific enough that a different evaluator would reach substantially similar conclusions. That transferability is the practical test of whether the methodology scales beyond one person.